Structural Description of the Collatz Inverse Tree via Mersenne-Fermat Factorization and Jacobsthal Sequences

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Abstract

In this report, the generation structure of the Inverse Tree in the Collatz conjecture is reinterpreted through Mersenne-type transformations, Fermat-type difference of squares factorization, and their correspondence with specific Jacobsthal (J) sequences. By transforming an odd nucleus a into the form $a \cdot 2^k - 1$ and embedding it into the difference of squares $(a \cdot 2^k)^2 - 1$, it splits into a Mersenne (M)-type factor and a Fermat (F)-type factor. The property that one of these two-body factors must be a multiple of 3 structurally matches the branching condition of the Collatz inverse tree. Furthermore, we show that the mod 3 oscillatory structure resonates with the sign oscillation of the J-series, and the germination condition of the inverse tree aligns perfectly with the parity structure of the J-series.

1. Collatz Inverse Tree and Germination Conditions for Odd Nuclei

The Collatz inverse mapping (inverse tree) is given by:

$$a_{\text{next}} = \frac{a \cdot 2^k - 1}{3}$$

Here, the condition for the next odd nucleus a_{next} (the previous odd nucleus in the forward mapping) to be an integer is:

$$a \cdot 2^k - 1 \equiv 0 \pmod 3$$

This study demonstrates that this condition naturally arises from the Mersenne-Fermat (M-F) factorization and the mod 3 oscillatory structure.

2. Mersenne-ization of Odd Nuclei and Fermat-type Difference of Squares Factorization

Convert the odd nucleus a into a Mersenne type and embed it into a difference of squares:

$$(a \cdot 2^k)^2 - 1 = (a \cdot 2^k - 1)(a \cdot 2^k + 1)$$

Here, we define the two-body factors after splitting as follows:

- $a \cdot 2^k - 1$: Mersenne (M)-type factor
- $a \cdot 2^k + 1$: Fermat (F)-type factor

When the odd nucleus a is not a multiple of 3 ($a \equiv \pm 1 \pmod 3$), the power of 2 dictates, via the following congruence condition, that one of the M-type or F-type factors must strictly be a multiple of 3.

$$2^k \equiv (-1)^k \pmod 3$$

This structurally coincides with the condition determining "which branch germinates" in the Collatz inverse tree.

3. Selection Structure via Jacobsthal Sequences

The Jacobsthal (J) sequence J_n is defined by the following linear recurrence relation:

$$J_{n+1} = J_n + 2J_{n-1}$$

Introducing a "conjugate J sequence (cJ)" that incorporates sign inversion naturally corresponds to the mod 3 oscillatory structure.

Due to the alternating mod 3 oscillation of the power of 2 and the mod 3 congruence class of the odd nucleus a, the M-type and F-type factors are sorted into either a triple J sequence (3J) or a cJ sequence, depending on the parity of the hierarchy level k, as shown below:

1) For $a \equiv 1 \pmod 3$ (e.g., $a = 1, 7, 13$)
   - Even k: M-type → 3J number, F-type → cJ number
   - Odd k: M-type → cJ number, F-type → 3J number

2) For $a \equiv -1 \pmod 3$ (e.g., $a = 5, 11, 17$)
   - Even k: M-type → cJ number, F-type → 3J number
   - Odd k: M-type → 3J number, F-type → cJ number

4. Resonance between Jacobsthal Sequences and the Collatz Inverse Tree

As an example, for $a = 5$ ($\equiv -1 \pmod 3$), the inverse tree germinates only when k is odd:

$$5 \cdot 2^k - 1 \equiv (-1)(-1)^k - 1 \equiv (-1)^{k+1} - 1 \pmod 3$$

The sequence of odd nuclei actually obtained is shown below:

$$3, 13, 53, 213, \ldots$$

This exactly matches the even terms of the J-type recurrence relation:

$$a_{n+1} = 2a_n + (-1)^n$$

$$J(2,3) = \{2, 3, 7, 13, 27, 53, 107, 213, \ldots\}$$

From the above, it is evident that the mod 3 oscillation and the sign structure of the J sequence are in resonance.

5. Verification Example: Collatz Inverse Tree of Initial Value 7

The initial value 7 possesses a typical descending route in the Collatz inverse tree, making it an optimal example to demonstrate consistency with the J-series.

$$1 \to 5 \to 13 \to 17 \to 11 \to 7$$

Checking the correspondence between the M-F factorization and the J-series for each a confirms that all perfectly align with either the J-series or the cJ-series.

5.1 $\boldsymbol{a = 1}$

- $k = 2, 4, 6, \ldots$
- $a_{\text{next}} = 1, 5, 21, \ldots$
- J-series: $J_k(0,1)$
- $k = 4$
- M-type factor: 15 (3J) → 5 (J)
- F-type factor: 17 (cJ)

5.2 $\boldsymbol{a = 5}$

- $k = 1, 3, 5, \ldots$
- $a_{\text{next}} = 3, 13, 53, \ldots$
- J-series: $J_k(2,3)$
- $k = 3$
- M-type factor: 39 (3J) → 13 (J)
- F-type factor: 41 (cJ)

5.3 $\boldsymbol{a = 13}$

- $k = 2, 4, 6, \ldots$
- $a_{\text{next}} = 17, 69, 277, \ldots$
- J-series:$J_k(4,9)$
- $k = 2$
- M-type factor: 51 (3J) → 17 (J)
- F-type factor: 53 (cJ)

5.4 $\boldsymbol{a = 17}$

- $k = 1, 3, 5, \ldots$
- $a_{\text{next}} = 11, 45, 181, \ldots$
- J-series: $J_k(6,11)$
- $k = 1$
- M-type factor: 33 (3J) → 11 (J)
- F-type factor: 35 (cJ)

5.5 $\boldsymbol{a = 11}$

- $k = 1, 3, 5, \ldots$
- $a_{\text{next}} = 7, 29, 117, \ldots$
- J-series: $J_k(4,7)$
- $k = 1$
- M-type factor: 21 (3J) → 7 (J)
- F-type factor: 23 (cJ)

Thus, taking the Collatz inverse tree of the initial value 7 as an example, the correspondence between the M-F factorization of each odd nucleus and the J-series is illustrated in **Figure 1**.

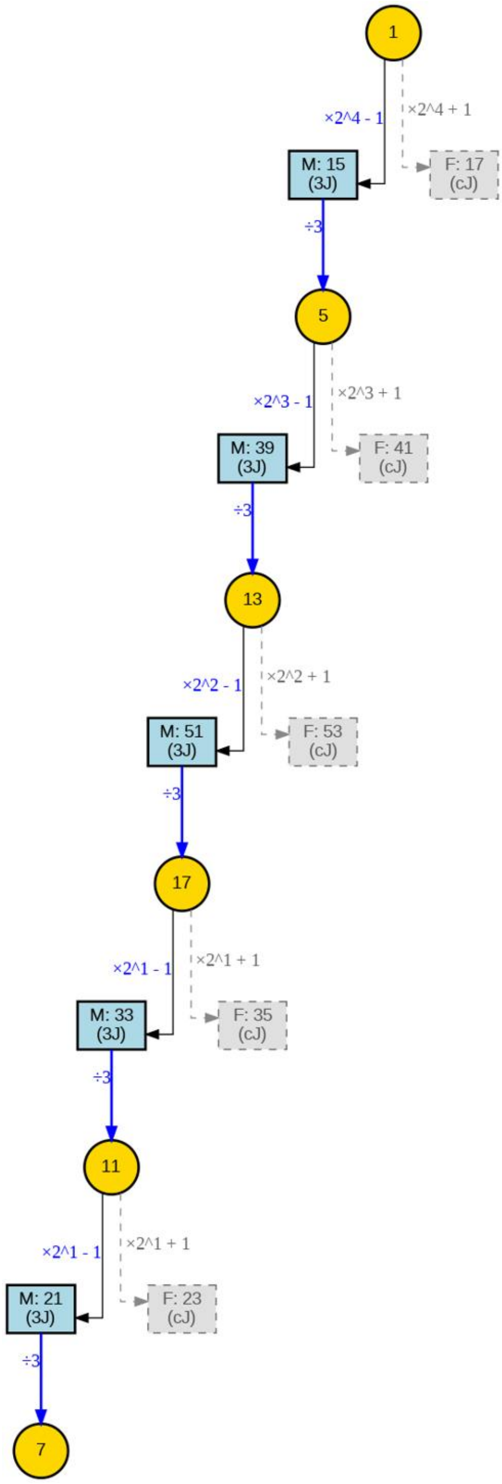

Figure 1. Mersenne-Fermat cascade decay of odd nuclei.
The diagram illustrates the reverse odd Collatz orbit for the initial value 7 ($1 \to 5 \to 13 \to 17 \to 11 \to 7$), demonstrating the separation of M-type (3J) and F-type (cJ) factors at each generation.

6. Conclusion

Observations from this study suggest the following:

- The Mersenne-ization and Fermat-type factorization of odd nuclei structurally coincide with the mod 3 branching condition of the Collatz inverse tree.
- The mod 3 oscillation of the M-type and F-type factors aligns with the J sequence and its conjugate J sequence.
- In specific examples, the sequence of odd nuclei in the inverse tree matches specific J-type recurrence relations.

These correspondences indicate the possibility that the structure of the Collatz inverse tree reflects the algebraic properties of the J-series.

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